

Caleb AI: Engineering Trust in the Verification Era

1. The Crisis of the Generative Era and the Shift to Verification

The "Generative Era" has hit its ceiling. While Large Language Models (LLMs) excel at high-level planning and synthesis, they remain fundamentally unreliable for exact computation and formal validation. As AI systems migrate from stochastic prototypes to production-grade technical strategies, we are entering the "Verification Era." This shift represents an architectural mandate: we must move away from undifferentiated model passes toward a governed system that separates adaptive reasoning from deterministic execution. The core doctrine of Caleb AI assumes that AI models are fallible. They require a deterministic substrate to provide logic-driven execution—tasks like arithmetic, unit conversion, and strict schema adherence—where models consistently fail. By restricting models to "adaptive reasoning" (resolving ambiguity and abstraction) and delegating mechanical labor to verified, local components, Caleb AI ensures system reliability. **The Caleb AI Doctrine Models think, Hollows work, Caleb orchestrates.**

2. Architectural Topology: The Five Planes of Control

Caleb AI is structured into five conceptual planes. Each maintains strict "non-responsibilities" to prevent logic leakage and ensure that reasoning never bypasses deterministic discipline.

- **Orchestration Core:** The policy brain. It owns task classification, workflow selection, and final assembly. It must not perform raw arithmetic, parsing, or static validation; it remains a control plane, never a labor plane. While the Core handles the policy of assembly, it delegates the materialization of artifacts to the **Forge**.
- **Model API Layer:** The adaptive reasoning surface. It owns planning, interpretation, and domain reasoning. It must not own narrow mechanical transforms or deterministic calculations. It treats the model as a selector of tools, not an executor of logic.
- **Communication Bus:** The transport layer for versioned envelopes. It owns contracts, trace IDs, and authorization. Following MCP (Model Context Protocol) standards, it manages lifecycle and capability negotiation. It must not own business logic or prompt policies, focusing strictly on transport integrity.
- **Hollow Server:** The deterministic execution engine. It owns repeatable local labor and inspection. It must not own open-ended judgment or final policy synthesis. It remains narrow, testable, and local-first to reduce round-trips and failure exposure.
- **Verified Return Path:** The mandatory gatekeeper. It owns the multi-gate verification pipeline and trust-tier assignment. It acts as the deciding checkpoint and must not own the initial generation of content, only its strict validation.

Caleb AI Component Reference Design

Component, Primary Responsibility, Source-Backed Precedent

Orchestration Core, "Routing, state management, and policy.", "LangGraph, AutoGen, CrewAI"

Model API Layer, "Interpretation, critique, and synthesis.", "Toolformer, ReAct"

Communication Bus, Capability/Lifecycle negotiation and integrity, "MCP, JSON-RPC 2.0"

Hollow Server, Local deterministic task execution., "PAL, Program of Thoughts"
Verified Return Path, Multi-gate output validation and trust-tiering., "NIST AI RMF, OWASP GenAI"

3. Deterministic Workloads: The Hollow Server and Local-First Execution

"Hollows" are deterministic, callable execution units for logic-driven work. Caleb AI adopts a "Local-First" rationale: keeping deterministic labor local reduces latency, improves privacy by keeping sensitive data out of provider logs, and removes exposure to remote API failures for stable logic.

Hollow Taxonomy

The architecture categorizes Hollows into specialized units:

- **Compute:** Arithmetic, financial math, and timeline scoring.
- **Validation:** JSON Schema checks and contradiction scans.
- **Transformation:** Normalization, parsing, and tabular shaping.
- **Policy:** Allow/deny gating and side-effect control.
- **Provenance:** Hashing, stamping, and ledger writing.
- **Routing:** Deterministic triage and workflow selection.
- **Recovery:** Retry/backoff and dead-letter handling (RFC 9457 semantics).

Recommended Initial Hollow Catalog

- **Calculator:** Handles exact financial math and units (Inputs: formulas; Outputs: exact results).
- **Schema Validator:** Strict adherence to JSON Schema 2020-12 (Inputs: JSON payload; Outputs: valid/invalid status).
- **Repo/AST Inspector:** Inspects code symbols and dependencies (Inputs: repository path; Outputs: symbol index, dependency tree).
- **Text Transform:** Normalizes text and performs slugification (Inputs: text blob; Outputs: cleaned structure).
- **Sandboxed Command:** Executes builds or ffmpeg transforms (Inputs: scoped command; Outputs: stdout/stderr, exit code).

4. The Verified Return Path: Policing the Model-Hollow Boundary

The Verified Return Path is the mandatory gate between raw output and trusted system state. No result influences synthesis unless it passes the following pipeline:

1. **Schema Validation:** Strict compliance with JSON Schema.
2. **Hash/Replay Confirmation:** Ensuring deterministic results are reproducible.
3. **Policy Evaluation:** Checking results against safety and compliance rubrics.
4. **Provenance Stamping:** Linking the result to its originating activity and agent.

The Trust Hierarchy

Artifacts are promoted through five tiers:

- **T0:** Raw model suggestion (Unverified).

- **T1:** Schema-valid but unverified output.
- **T2:** Deterministic Hollow output (Reproduced or hash-confirmed).
- **T3:** Deterministic output with policy clearance and provenance anchoring.
- **T4:** Human-approved or externally authoritative result with full citation chain.

Hollow Return Envelope

Every Hollow invocation must return a metadata-rich envelope as defined in the Caleb AI specification:

```
{
  "schema_version": "1.2.0",
  "trace_id": "trc_01JX...",
  "run_id": "run_01JX...",
  "pass_id": "pass_analyst_02",
  "call_id": "call_9f8f...",
  "issued_by": {
    "role": "Analyst",
    "model": "vendor:model-name",
    "prompt_hash": "sha256:..."
  },
  "status": "ok",
  "validation": {
    "schema_valid": true,
    "content_hash": "sha256:...",
    "warnings": []
  },
  "timing": {
    "started_at": "2026-06-06T14:08:00Z",
    "ended_at": "2026-06-06T14:08:01Z",
    "duration_ms": 921
  }
}
```

5. Advanced Reasoning: Conditional Role Rotation and Handoff Gates

Role Rotation is an escalation policy for high-stakes or ambiguous tasks. It is invoked only when simpler direct paths fail or risk crosses a threshold.

Rotating Roles and Responsibilities

- **Planner:** Task decomposition, evidence requirements, and decomposition graphs.
- **Analyst:** Evidence gathering via Hollows and construction of candidate findings.
- **Critic:** The "Role Handoff Gate." Challenges assumptions, identifies contradictions, and flags schema failures.
- **Synthesizer:** Final integration and trust-tier assignment.

Escalation Triggers

The Role Router moves from a direct path to a rotated path based on:

- **Ambiguity:** Request lacks clear deterministic boundaries.
- **High Stakes:** Task involves consequential actions or policy-sensitive data.
- **Multi-step Dependency:** Requires a complex graph of sub-tasks.
- **Low Consensus:** Disagreement between model outputs or tool results (a cheaper alternative to rotation via majority voting).

6. Cryptographic Provenance: The Ledger as the System of Record

The Ledger is an append-only system utilizing W3C PROV-O concepts. It links **Entities** (artifacts), **Activities** (invocations), and **Agents** (Core/Model) to ensure full "Replayability." The **Verified Return Path Rule** mandates that no result influences synthesis without a linked record in the Ledger.

Ledger Entry Metadata (PROV-O Mapping)

```
{
  "prov:Activity": "run_01JZC8Z6M0S4Y2QJ7X4M8W2G9A",
  "prov:Agent": "caleb-core-v1",
  "prov:used": ["artifact://candidate/output.json"],
  "prov:generated": ["artifact://verified/output.json"],
  "input_hash": "sha256:3d5d...a2",
  "output_hash": "sha256:9aa1...42",
  "schema_version": "1.2.0",
  "trust_tier": "T2",
  "timestamp": "2026-06-06T14:08:01Z"
}
```

7. Security, Governance, and Sandboxing

Caleb AI maintains hard trust boundaries at the prompt, Hollow, and filesystem levels.

- **Least Privilege:** Hollows use capability-scoped handles (opaque directory jail rules) instead of raw paths.
- **Sandboxed Execution:** Any Hollow running commands (e.g., Docker) operates with default-deny network egress.
- **Approval Boundaries:** Side-effecting Hollows that alter external state require explicit authorization before actuation.
- **Mitigations:** These controls address OWASP GenAI risks (prompt injection, excessive agency) by ensuring untrusted suggestions never translate to action without a deterministic gate.

8. The Economics of Verification: Performance and Cost Optimization

Caleb AI rejects "Universal Deep Reasoning" in favor of hybrid orchestration.

- **Prompt Caching:** Placing stable orchestration context and Hollow manifests at the start of the prompt reduces latency by up to 80% and costs by up to 90%.
- **Hollow Efficiency:** Replacing expensive model tokens with local CPU execution for deterministic tasks provides massive economic scaling.

Practical Metrics for the Verification Era

- **Hollow Hit Rate:** Percentage of work handled by deterministic units.
- **Direct-Path Rate:** Frequency of tasks avoiding the high-cost rotation path.
- **Validation Disagreement Rate:** Frequency of model/Hollow contradictions.
- **Replay Completeness:** Success rate of reconstructing runs via the Ledger.

9. Implementation Roadmap: A Phased Deployment Strategy

Phase 1: The Core (Wave 1)

Establish the Orchestration Core and a minimal catalog of 5-10 Hollows (Calculator, Validator, Repo Inspector). Implement the internal request/response envelope and Ledgered replay.

Phase 2: Dynamic Routing (Wave 2)

Implement a richer Hollow catalog and a provider-agnostic Communication Bus using MCP. Resolve manifests outside the primary model prompt to minimize context bloat.

Phase 3: Advanced Reasoning (Wave 3)

Deploy conditional role rotation for high-value tasks. Implement cross-model variants for adversarial checking and consensus-based routing.

Concise Implementation Checklist

- Define internal request/response envelope (trace_id, pass_id, etc.).
- Implement strict JSON Schema 2020-12 validation at the bus boundary.
- Build initial Hollow catalog for arithmetic, symbols, and validation.
- Establish append-only Ledger utilizing W3C PROV concepts.
- Configure isolated Docker sandboxes for side-effecting Hollows.

10. Conclusion: The Future of Governed AI Systems

Caleb AI is a refinement of modern agent design, not a rejection of it. By providing an engineering-grade substrate for language models, it delivers strategic advantages in reliability, auditability, and economic sustainability. By separating reasoning from execution and enforcing a verified return path, the architecture ensures that AI systems are not just capable, but governed. **Caleb AI is the practical synthesis of adaptive reasoning and deterministic discipline.**